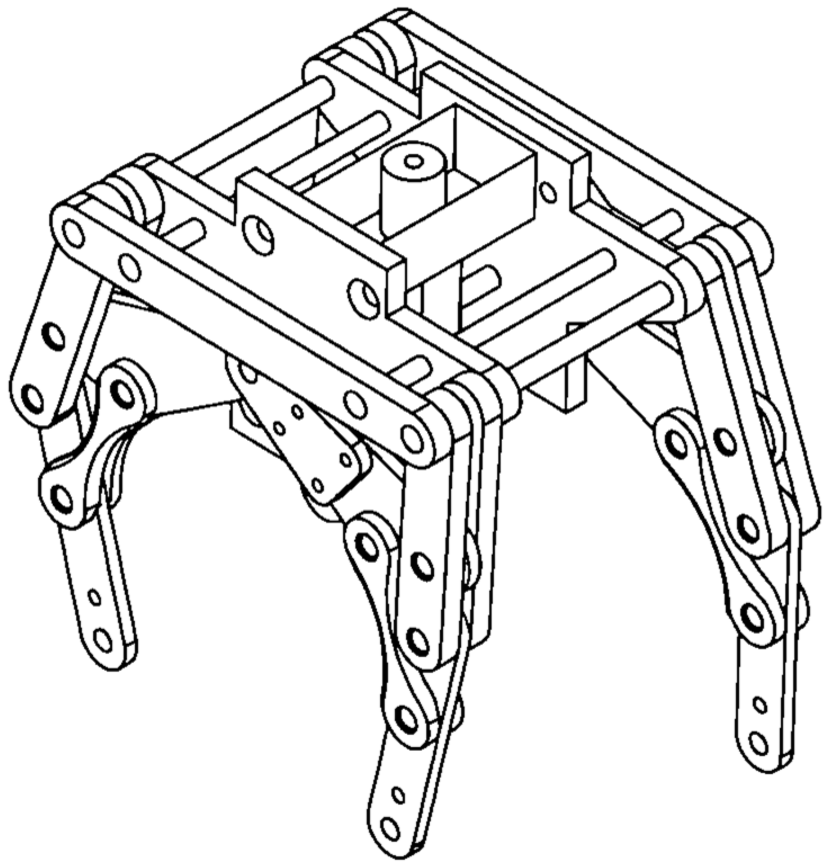
Quadruped Dog Robot



1. **Body & Frame Shape**
   The structure consists of a rigid central frame (chassis) that houses the motor and transmission components in the middle, with lateral shafts extending outward to support the legs.

2. **Leg Design and Motion Mechanism**
   The legs do not rely on independently controlled joints. Instead, they use a linkage mechanism.

3. **Number of Joints and Degrees of Freedom (DoF)**
   Despite having many pin joints connecting the links, the robot effectively possesses only 1 Degree of Freedom per side. This is because the movement of all legs is driven by a single rotational source.

4. **Motor Selection**
   You do not need expensive servo motors to track precise angles. Instead, a robust DC Geared Motor or a Continuous Rotation Servo is ideal.

5. **Initial Torque Calculation for a Joint**
   To calculate the initial torque required for a joint in this linkage mechanism, you must focus on the worst-case scenario during the robot's stance phase when the legs support their entire weight. The total torque is computed using the formula $T_{total} = m \cdot g \cdot L_{max}$, where m is the total mass of the robot (e.g., 2 kg), g is the acceleration due to gravity (9.81 m/s^2), and L_{max} is the maximum horizontal distance from the central driving shaft to the foot's point of contact with the ground during its stride. Because this central motor drives multiple linkages simultaneously, the resulting value must be multiplied by a safety factor of at least 1.5 to 2 to effectively overcome the heavy mechanical friction and backlash accumulated across the numerous sequential pin joints.

6. **Stability and Center of Gravity (CoG)**
   The robot relies on Static Stability. To prevent the robot from tipping over while walking, the projection of its Center of Gravity must always fall within the support polygon formed by the feet currently touching the ground.

7. **Proposed Gait Strategy**
   The walking gait is hardwired into the mechanical design based on the assembly angles of the linkages on the power takeoff shaft.

8. **Anticipated Mechanical Issues**

   - Backlash and Joint Clearance: Having many sequential joints means that even a tiny amount of play (loose tolerance) in each joint will accumulate. This can cause the legs to become "wobbly" and unstable when hitting the ground.
   - High Friction: More moving parts mean more friction, which drains motor torque. You will need to use small bearings or manufacture the joints with high precision (using 3D printing or laser cutting).
   - Steering Limitations: If a single motor drives both sides simultaneously, the robot can only walk in a straight line. To enable steering, it is best to separate the left and right sides, powering each side with its own independent motor.